MSG No. 89.94 P_I422 [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `422`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `422`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,10,11,12,13,14,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `222.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,4,5,6]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[2,3,7,8]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[9,12,13,14]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[10,11,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,6,15,16]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[2,3,12,13]`
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[4,5,10,11]`
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[7,8,9,14]`

Table 5: Wyckoff site: `4e`, site symmetry: `4..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,6]`
2	$[0,\ 0,\ -z]$	`[4,5,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,10,11,14]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[12,13,15,16]`

Table 6: Wyckoff site: `8f`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,6]`
2	$\left[\frac{1}{2},\ 0,\ z\right]$	`[2,3]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[4,5]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[7,8]`
5	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[9,14]`
6	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,11]`
7	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[12,13]`
8	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[15,16]`

Table 7: Wyckoff site: `8g`, site symmetry: `..2`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,7]`
2	$[-x,\ x,\ 0]$	`[2,5]`
3	$[x,\ -x,\ 0]$	`[3,4]`
4	$[-x,\ -x,\ 0]$	`[6,8]`
5	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[9,15]`
6	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[10,13]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[11,12]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[14,16]`

Table 8: Wyckoff site: `8h`, site symmetry: `.2.`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,4]`
2	$[0,\ x,\ 0]$	`[2,7]`
3	$[0,\ -x,\ 0]$	`[3,8]`
4	$[-x,\ 0,\ 0]$	`[5,6]`
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,12]`
6	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[10,15]`
7	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[11,16]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[13,14]`

Table 9: Wyckoff site: `8i`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,4]`
2	$\left[0,\ x,\ \frac{1}{2}\right]$	`[2,7]`
3	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[3,8]`
4	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[5,6]`
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,12]`
6	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[10,15]`
7	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[11,16]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	`[13,14]`

Table 10: Wyckoff site: `8j`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, \frac{1}{4}\right]$	`[1,15]`
2	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{4}\right]$	`[2,13]`
3	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{4}\right]$	`[3,12]`
4	$\left[x,\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	`[4,11]`
5	$\left[-x,\, x+\frac{1}{2},\, \frac{3}{4}\right]$	`[5,10]`
6	$\left[-x,\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	`[6,16]`
7	$\left[x+\frac{1}{2},\, x,\, \frac{3}{4}\right]$	`[7,9]`
8	$\left[\frac{1}{2}-x,\, -x,\, \frac{3}{4}\right]$	`[8,14]`

Table 11: Wyckoff site: `16k`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[-y,\, x,\, z]$	`[2]`
3	$[y,\, -x,\, z]$	`[3]`
4	$[x,\, -y,\, -z]$	`[4]`
5	$[-x,\, y,\, -z]$	`[5]`
6	$[-x,\, -y,\, z]$	`[6]`
7	$[y,\, x,\, -z]$	`[7]`
8	$[-y,\, -x,\, -z]$	`[8]`
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[9]`
10	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[10]`
11	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[11]`
12	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	`[12]`
13	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[13]`
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	`[14]`
15	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[15]`
16	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	`[16]`